

Stochastic M/M/c Erlang C Congestion Modeling & Closed-Form Tail Inversion Architecture

Continuous-Time Markov Chain State Space --> Erlang C Congestion --> Analytic Inversion --> Theorem 1 Invariant Order

1. PARAMETER CONDITIONING & GUARD

Poisson Arrival Process

Mean Arrival Rate: λ (vehicles/hour)

Non-homogeneous Poisson: $\lambda(t) = \tilde{\lambda} \cdot \delta_{\text{diurnal}}(t)$

Aggregated spatial POI & commuter flux

λ, μ, c

Multiserver Topology & Service

Parallel Charging Ports: $c \in \{2, 4, 8, 16\}$

Service Rate per Port: $\mu = 2.0$ veh/h

Mean Charging Duration: $1/\mu = 30.0$ min

Exponential Service Distribution: $B(t) = 1 - e^{-\mu t}$

Offered Load & Server Intensity

Offered Traffic Load: $a = \frac{\lambda}{\mu}$ (Erlangs)

Mean Server Utilization: $\rho = \frac{\lambda}{c \cdot \mu} = \frac{a}{c}$

Critical Traffic Threshold: $\rho \rightarrow 1.0^-$

Under-saturated vs. congested boundary

Ergodicity & Stability Guardrail

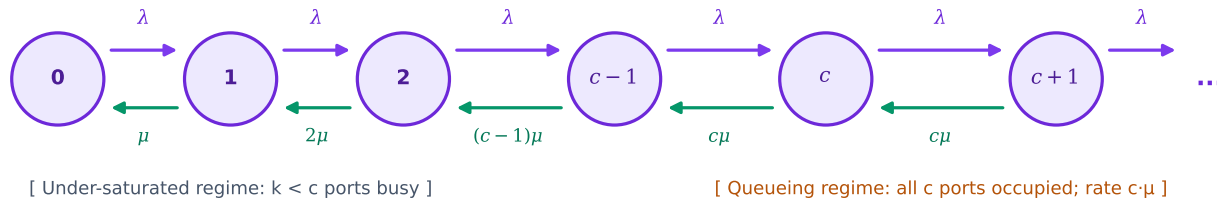
Stability Condition: $\rho = \frac{\lambda}{c\mu} < 1.0$

Prevents infinite queue divergence
Guarantees stationary state distribution π

Circuit Breaker: If $\rho \geq 1.0 \Rightarrow$ Outage Flag
Triggers load shedding & dynamic re-routing

2. CONTINUOUS-TIME MARKOV CHAIN (CTMC) & ERLANG C SOLVER

Birth-Death Transition Lattice: State Space $\{0, 1, \dots, c-1, c, c+1, \dots\}$



CDF $P(W_q \leq t)$

Closed-Form Erlang C Congestion & Waiting Time CDF

Delay Probability (All Ports Saturated):

$$C(c, a) = P(W_q > 0) = \frac{\frac{a^c}{c!(1-\rho)}}{\sum_{k=0}^{c-1} \frac{a^k}{k!} + \frac{a^c}{c!(1-\rho)}}$$

Where $a = \lambda/\mu$, $\rho = a/c < 1.0$

Defective Exponential Distribution:

$$P(W_q \leq t) = 1 - C(c, a) e^{-c\mu(1-\rho)t}, \quad t \geq 0$$

Point mass at zero wait: $P(W_q = 0) = 1 - C(c, a)$

Continuous tail for $t > 0$ under queueing condition

Delay Probability $C(c, a)$

3. EXACT TAIL PERCENTILE INVERSION & SPEEDUP ACCELERATOR

Closed-Form Quantile Inversion Solver

$$\text{Set } P(W_q \leq t_p) = p \Rightarrow 1 - C(c, a) e^{-c\mu(1-\rho)t_p} = p$$

$$t_p = \max\left(0, -\frac{\ln\left(\frac{1-p}{C(c, a)}\right)}{c\mu(1-\rho)}\right) = \max\left(0, \frac{\ln\left(\frac{C(c, a)}{1-p}\right)}{c\mu(1-\rho)}\right)$$

Median Wait Time (p_{50} , $p = 0.50$):

$$t_{0.50} = \max\left(0, \frac{\ln(2 \cdot C(c, a))}{c\mu(1-\rho)}\right)$$

Tail Wait Time (p_{90} , $p = 0.90$):

$$t_{0.90} = \max\left(0, \frac{\ln(10 \cdot C(c, a))}{c\mu(1-\rho)}\right)$$

t_{50}, t_{90}

Empirical Computational Benchmark vs. Discrete Event Simulation (DES)

Analytic Inversion Latency

< 0.15 ms

Closed-form $O(c)$ evaluation

DES Monte Carlo Latency

15.2 seconds

25,000 session trajectories

Real-Time Acceleration

> 100,000x Speedup

Zero simulation barrier

4. THEOREM 1 INVARIANT VERIFICATION

Theorem 1: Monotonic Ordering Invariant

$$\forall \rho < 1.0, \forall c \in \mathbb{N}^+ : t_{0.50} \leq t_{0.90}$$

Regime 1: $C(c, a) \leq 0.10$

--> $t_{0.50} = 0 \leq t_{0.90} = 0$

Point mass domin.: zero wait at both

Regime 2: $0.10 < C(c, a) \leq 0.50$

--> $t_{0.50} = 0 < t_{0.90}$

Median zero; tail wait emerges

Regime 3: $C(c, a) > 0.50$

--> $0 < t_{0.50} < t_{0.90}$

$\ln(2C) < \ln(10C) \Rightarrow$ Strict order

Empirical Validation Across 728 Regimes

Total Regimes Tested

728 distinct (λ, μ, c)

Monotonicity Violations

0 Violations (100.0% Adherence)

Ground Truth MAE (vs. DES)

9.39 minutes

Ground Truth RMSE

15.72 minutes

Downstream Multi-Agent Delivery

Provenance Contract: [ESTIMATED]
Confidence Score: 0.70 - 0.94

Feeds ScoringAgent (Wait Utility Penalty)
Feeds ExplanationAgent (Gemini 2.0 Hedging)
Guarantees zero negative tail delays

FastAPI response: sub-second delivery